



Day 3

Synchronous Learning

AI-900 Certification

Preparation

Implementing Partner:



Daftar Materi

1. Introduction to Computer Vision Concepts
2. Get Started with Computer Vision in Azure
3. Introduction to AI-powered Information Extraction Concepts
4. Get Started with AI-powered Information Extraction in Azure

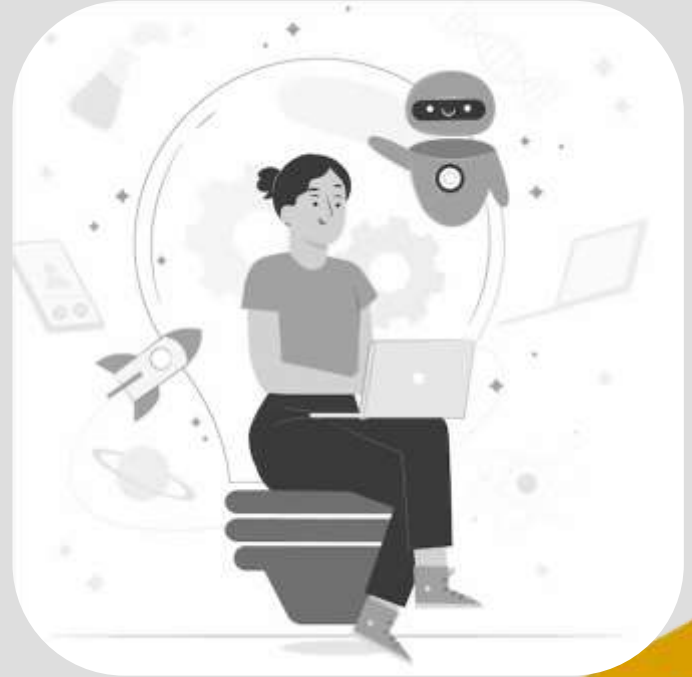


Bab 1

Introduction to Computer Vision Concepts

Daftar Isi

1. Overview
2. Understand Image Processing
3. Machine Learning for Computer Vision
4. Understand Modern Vision Models



Bab 1

Sub-bab 1

Overview





Computer Vision Capabilities

Type	Description
Image analysis	The ability to detect, classify, caption, and generate insights.
Spatial analysis	The ability to understand people's presence and movements within physical areas in real time.
Facial recognition	The ability to recognize and verify human identity.
Optical character recognition (OCR)	The ability to extract printed and handwritten text from images with varied languages and writing styles.

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Understand Image Processing



Image as Pixel Arrays

The array consists of seven rows and seven columns, representing the pixel values for a 7x7 pixel image (which is known as the image's *resolution*). Each pixel has a value between 0 (black) and 255 (white); with values between these bounds representing shades of gray. The image represented by this array looks similar to the following (magnified) image:

0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	255	255	255	0	0
0	0	255	255	255	0	0
0	0	255	255	255	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0

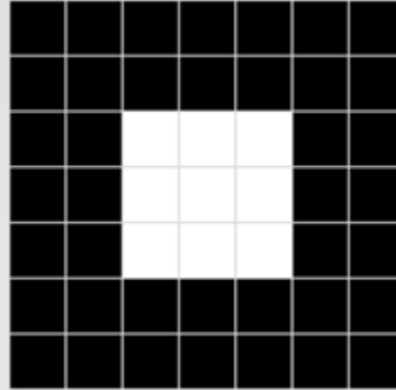
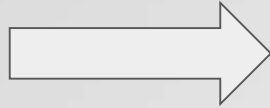


Image as Pixel Arrays

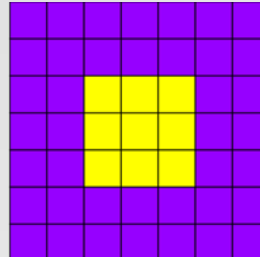
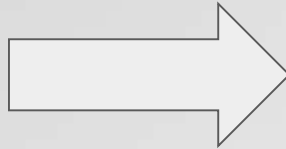
The array of pixel values for this image is two-dimensional (representing rows and columns, or x and y coordinates) and defines a single rectangle of pixel values. A single layer of pixel values like this represents a grayscale image. In reality, most digital images are multidimensional and consist of three layers (known as *channels*) that represent red, green, and blue (RGB) color hues. For example, we could represent a color image by defining three channels of pixel values that create the same square shape as the previous grayscale example:

```

Red:
150 150 150 150 150 150 150
150 150 150 150 150 150 150
150 150 255 255 255 150 150
150 150 255 255 255 150 150
150 150 255 255 255 150 150
150 150 150 150 150 150 150
150 150 150 150 150 150 150

Green:
0 0 0 0 0 0 0
0 0 0 0 0 0 0
0 0 255 255 255 0 0
0 0 255 255 255 0 0
0 0 255 255 255 0 0
0 0 0 0 0 0 0
0 0 0 0 0 0 0

Blue:
255 255 255 255 255 255 255
255 255 255 255 255 255 255
255 255 0 0 0 255 255
255 255 0 0 0 255 255
255 255 0 0 0 255 255
255 255 255 255 255 255 255
255 255 255 255 255 255 255
  
```





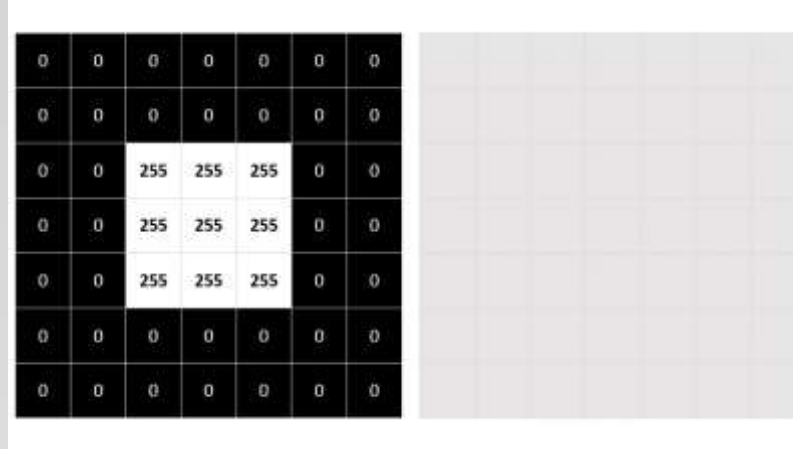
Filters

A common way to perform image processing tasks is to apply *filters* that modify the pixel values of the image to create a visual effect. A filter is defined by one or more arrays of pixel values, called filter *kernels*. For example, you could define filter with a 3x3 kernel as shown in this example:

```
-1 -1 -1  
-1 8 -1  
-1 -1 -1
```

The kernel is then *convolved* across the image, calculating a weighted sum for each 3x3 patch of pixels and assigning the result to a new image.

Convolutional Filtering



The filter is convolved across the image, calculating a new array of values. Some of the values might be outside of the 0 to 255 pixel value range, so the values are adjusted to fit into that range. Because of the shape of the filter, the outside edge of pixels isn't calculated, so a padding value (usually 0) is applied. The resulting array represents a new image in which the filter has transformed the original image. In this case, the filter has had the effect of highlighting the *edges* of shapes in the image.

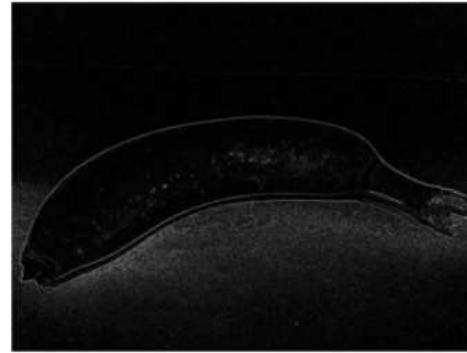
Convolutional Filtering

Because the filter is convolved across the image, this kind of image manipulation is often referred to as *convolutional filtering*. The filter used in this example is a particular type of filter (called a *laplace* filter) that highlights the edges on objects in an image. There are many other kinds of filter that you can use to create blurring, sharpening, color inversion, and other effects.

Original Image



Filtered Image



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Machine Learning for Computer Vision





Introduction

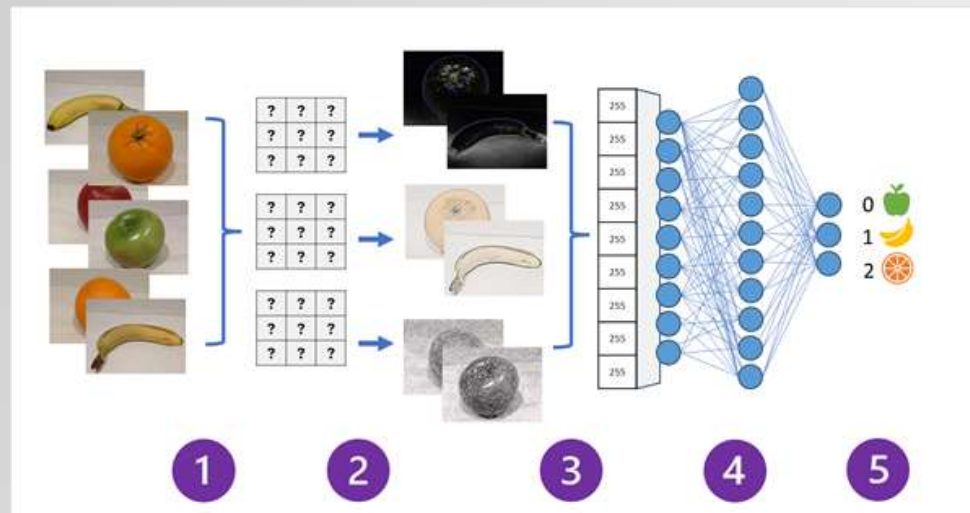
The ability to use filters to apply effects to images is useful in image processing tasks, such as you might perform with image editing software. However, the goal of computer vision is often to extract meaning, or at least actionable insights, from images; which requires the creation of machine learning models that are trained to recognize features based on large volumes of existing images.



Convolutional Neural Networks (CNN)

One of the most common machine learning model architectures for computer vision is a *convolutional neural network* (CNN), a type of deep learning architecture. CNNs use filters to extract numeric feature maps from images, and then feed the feature values into a deep learning model to generate a label prediction. For example, in an *image classification* scenario, the label represents the main subject of the image (in other words, what is this an image of?). You might train a CNN model with images of different kinds of fruit (such as apple, banana, and orange) so that the label that is predicted is the type of fruit in a given image.

During the *training* process for a CNN, filter kernels are initially defined using randomly generated weight values. Then, as the training process progresses, the models predictions are evaluated against known label values, and the filter weights are adjusted to improve accuracy. Eventually, the trained fruit image classification model uses the filter weights that best extract features that help identify different kinds of fruit.



1. Images with known labels (for example, 0: apple, 1: banana, or 2: orange) are fed into the network to train the model.
2. One or more layers of filters is used to extract features from each image as it is fed through the network. The filter kernels start with randomly assigned weights and generate arrays of numeric values called *feature maps*.
3. The feature maps are flattened into a single dimensional array of feature values.
4. The feature values are fed into a fully connected neural network.
5. The output layer of the neural network uses a *softmax* or similar function to produce a result that contains a probability value for each possible class, for example [0.2, 0.5, 0.3].

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Understand Modern Vision Models





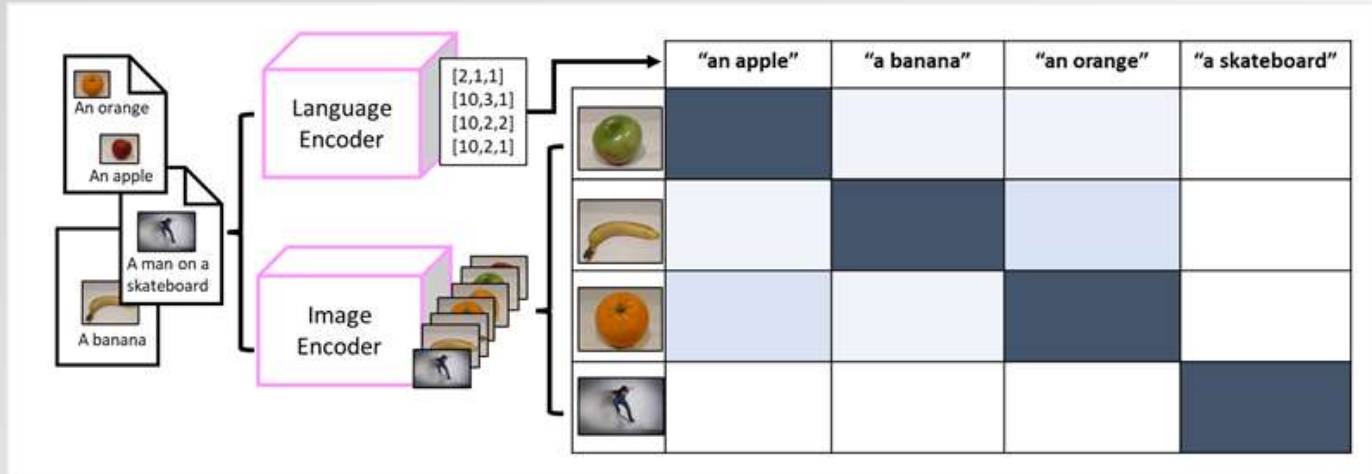
Introduction

CNNs have been at the core of computer vision solutions for many years. While they're commonly used to solve image classification problems as described previously, they're also the basis for more complex computer vision models. For example, *object detection* models combine CNN feature extraction layers with the identification of *regions of interest* in images to locate multiple classes of object in the same image.

But, how about transformers?

Multi-Modal Models

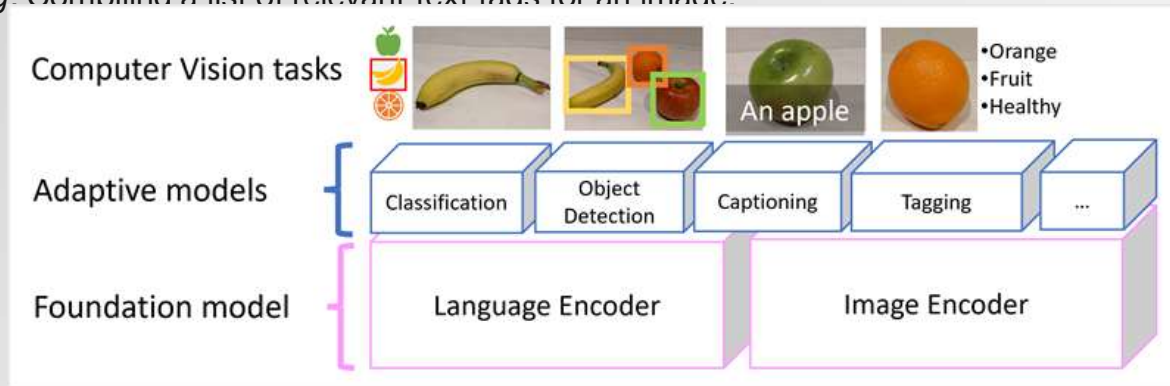
The success of transformers as a way to build language models has led AI researchers to consider whether the same approach would be effective for image data. The result is the development of *multi-modal* models, in which the model is trained using a large volume of captioned images, with no fixed *labels*. An image encoder extracts features from images based on pixel values and combines them with text embeddings created by a language encoder. The overall model encapsulates relationships between natural language token embeddings and image features, as shown here:



Bringing it all together

Modern vision models are trained with huge volumes of captioned images from the internet and include both a language encoder and an image encoder. Often users will interact with and adapted *foundation* models. Foundation models are pre-trained general models on which you can build multiple *adaptive* models for specialist tasks. For example, you can adapt a foundation model to perform:

- *Image classification*: Identifying to which category an image belongs.
- *Object detection*: Locating individual objects within an image.
- *Captioning*: Generating appropriate descriptions of images.
- *Tagging*: Compiling a list of relevant text tags for an image.



Akhir dari Bab 1

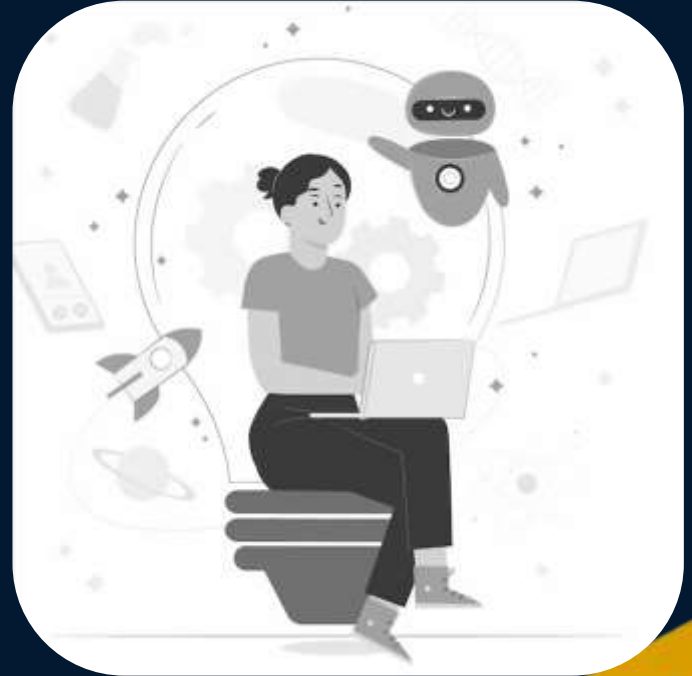


Bab 2

**Get Started with Computer Vision in
Azure**

Daftar Isi

1. Introduction
2. Understand Azure AI Services for Computer Vision
3. Understand Azure AI Vision Image Analysis Capabilities
4. Understand Azure AI Vision's Face service capabilities
5. Get Started in Azure AI Foundry Portal



Sub-bab 1

Introduction



Introduction

Computer vision is a field of artificial intelligence (AI) that enables machines to interpret and understand visual information from the world—such as images, videos, and live camera feeds. Computer vision capabilities support the automation of time-intensive tasks and enable possibilities that did not exist before.

Consider some of these applications of computer vision:

- *Manufacturing – Defect Detection:* AI vision systems inspect products on assembly lines in real time. They detect surface defects, misalignments, or missing components using object detection and image segmentation, reducing waste and improving quality control.
- *Healthcare – Medical Imaging Analysis:* Computer vision helps radiologists analyze X-rays, MRIs, and CT scans. AI models can highlight anomalies like tumors or fractures, assist in early diagnosis, and reduce human error.
- *Retail – Shelf Monitoring:* Retailers use AI vision to monitor store shelves. Cameras detect when products are out of stock or misplaced, enabling real-time inventory updates and improving customer experience.
- *Transportation – Autonomous Vehicles:* Self-driving cars rely on computer vision to recognize road signs, lane markings, pedestrians, and other vehicles. This enables safe navigation and decision-making in dynamic environments.

Sub-bab 2

Understand Azure AI Services for Computer Vision





Azure AI Vision

Azure AI provides a wide range of cloud-based services for various AI tasks, including computer vision. Microsoft's Azure AI Vision service provides **prebuilt and customizable computer vision models** that are based on deep learning models and provide various capabilities. Azure AI Vision provides "off-the-shelf" functionality for many common computer vision scenarios, while retaining the ability to create custom models using your own images.

Azure AI Vision Image Analysis service: Detects common objects in images, tags visual features, generates captions, and supports optical character recognition (OCR).

Azure AI Face service: Detects, recognizes, and analyzes human faces in images. Provides specific models for facial analysis that extend beyond what is available with image analysis.



Azure AI Vision Image Analysis Capabilities

Azure AI Vision's image analysis capabilities can be used with or without customization. Some of the capabilities that do not require customization include:

- Describing an image with captions
- Detecting common objects in an image
- Tagging visual features
- Optical character recognition

Input Image



Describing an image with

A person jumping on a skateboard

Detecting common objects in an image



- Skateboard (90.40%)
- Person (95.5%)

Tagging visual features

- sport (99.60%)
- person (99.56%)
- footwear (98.05%)

Optical Character Recognition (OCR)

Azure AI Vision service can use optical character recognition (OCR) capabilities to detect text in images. For example, consider the following image of a nutrition label on a product in a grocery store:



```
Nutrition Facts Amount Per Serving
Serving size:1 bar (40g)
Serving Per Package: 4
Total Fat 13g
Saturated Fat 1.5g
Amount Per Serving
Trans Fat 0g
calories 190
Cholesterol 0mg
ories from Fat 110
Sodium 20mg
ntDaily Values are based on
Vitamin A 50
calorie diet
```



Training Custom Model

If the built-in models provided by Azure AI Vision don't meet your needs, you can use the service to train a custom model for *image classification* or *object detection*. Azure AI Vision builds custom models on the pre-trained foundation model, meaning that you can train sophisticated models by using relatively few training images.

Sub-bab 3

Understand Azure AI Vision's Face Service Capabilities



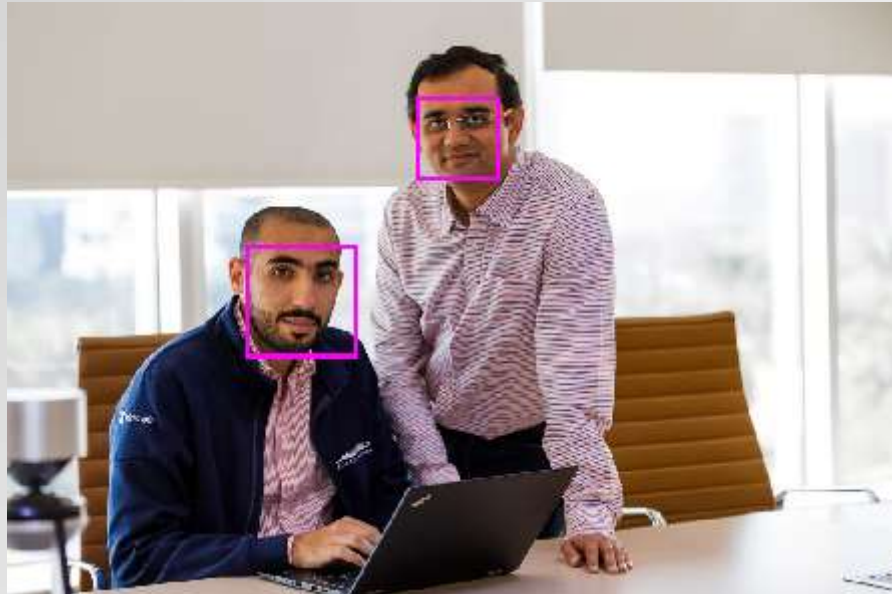


Introduction

As a product within Azure AI Vision, Azure AI Face supports specific use cases such as verifying user identity, liveness detection, touchless access control, and face redaction. Several concepts, including face detection and recognition, are essential to working with Face.

Facial Detection

Face detection involves identifying regions of an image that contain a human face, typically by returning *bounding box* coordinates that form a rectangle around the face, like this:



Facial Recognition

A further application of facial analysis is to train a machine learning model to identify known individuals from their facial features. This is known as *facial recognition*, and uses multiple images of an individual to train the model. This trains the model so that it can detect those individuals in new images on which it wasn't trained.





Responsible AI

Anyone can use the Face service to:

- Detect the location of faces in an image.
- Determine if a person is wearing glasses.
- Determine if there's occlusion, blur, noise, or over/under exposure for any of the faces.
- Return the head pose coordinates for each face in an image.

The Limited Access policy requires customers to [submit an intake form](#) to access additional Azure AI Face service capabilities including:

- Face verification: the ability to compare faces for similarity.
- Face identification: the ability to identify named individuals in an image.
- Liveness detection: the ability to detect and mitigate instances of recurring content and/or behaviors that indicate a violation of policies (e.g., such as if the input video stream is real or fake).

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Get Started in Azure AI Foundry Portal





Get Started

Azure AI Vision provides the building blocks for incorporating vision capabilities into applications. As one of many Azure AI services, you can create solutions with Azure AI Vision in several ways including:

- The Azure AI Foundry portal
- A software development kit (SDK) or REST API



Resources

To use Azure AI Vision, you need to create a resource for it in your Azure subscription. You can use either of the following resource types:

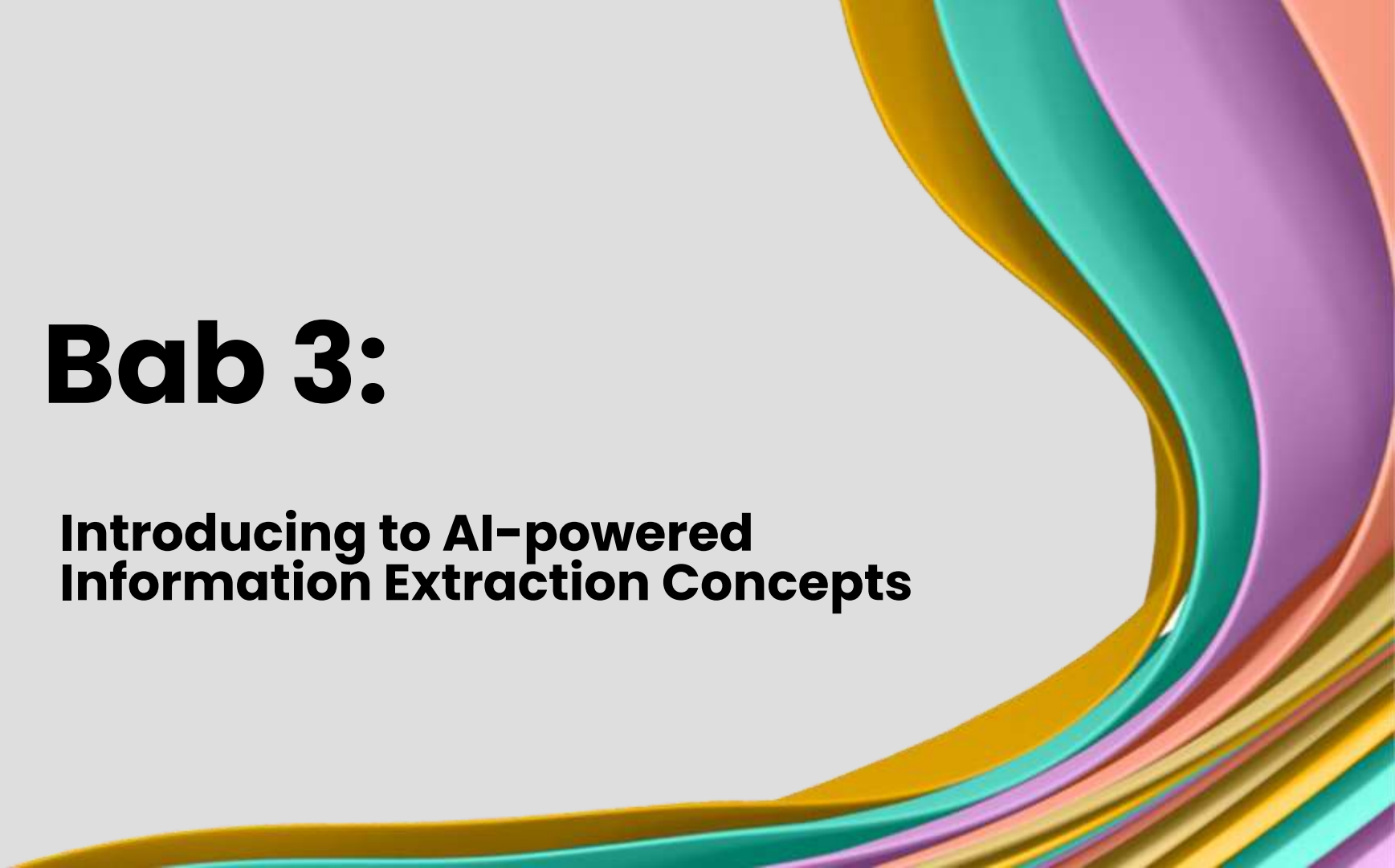
- **Azure AI Vision:** A specific resource for the Azure AI Vision service. Use this resource type if you don't intend to use any other Azure AI services, or if you want to track utilization and costs for your Azure AI Vision resource separately.
- **Azure AI services:** A general resource that includes Azure AI Vision along with many other Azure AI services; such as Azure AI Language, Azure AI Custom Vision, Azure AI Translator, and others. Use this resource type if you plan to use multiple AI services and want to simplify administration and development.

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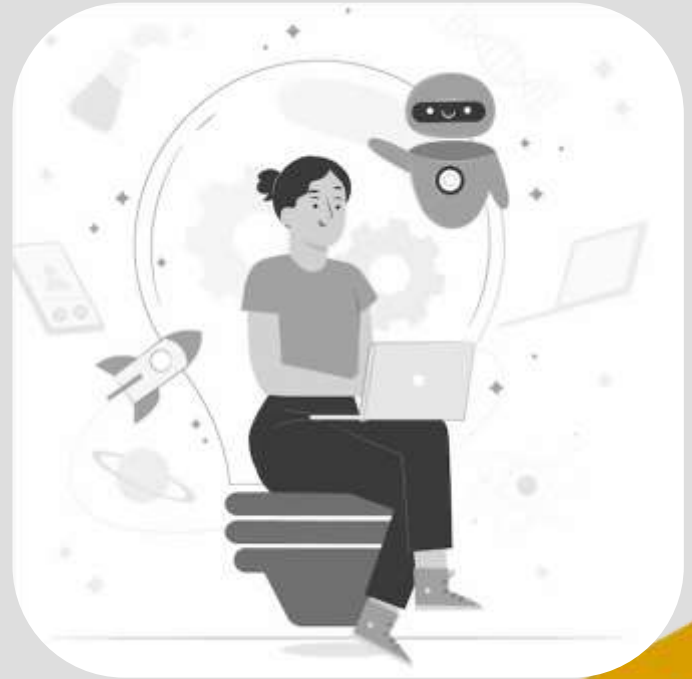
Bab 3:

**Introducing to AI-powered
Information Extraction Concepts**



Daftar Isi

1. Overview
2. Understand the extraction of data from images
3. Understand the extraction of data from forms
4. Understand multimodal data extraction
5. Understand data extraction for knowledge mining



Sub-bab 1

Overview





Introduction

Today's organizations deal with all kinds of content such as documents, video, audio, images, and text. A common task in these organizations includes identifying and storing key information from the content into databases.

Consider some of these use cases:

- A manufacturer has images of each of its products. The images need to be analyzed for defects and anomalies.
- A business works with a high volume of invoices, contracts, and reports with charts. Key data and summaries from the documents need to be extracted and logged.
- Many hours of customer calls are recorded for quality purposes. The audio needs to be transcribed, summarized, and analyzed for sentiment.
- A streaming catalog contains a large volume of video. Important moments in each video need to be tagged with metadata based on their content.

Manually processing such content can be slow and potentially error-prone. AI-powered information extraction encompasses capabilities that extract meaning from content. In this module, you explore core concepts related to information extraction.



Information Extraction Processes

Step	Description
Source Identification	Determine where the information resides and if it needs to be digitized.
Extraction	Leverages many techniques based on machine learning to understand and extract data from digitized content.
Transformation & Structuring	Extracted data is transformed into structured formats like JSON or tables.
Storage & Integration	The processed data is then stored in databases, data lakes, or analytics platforms for further use.

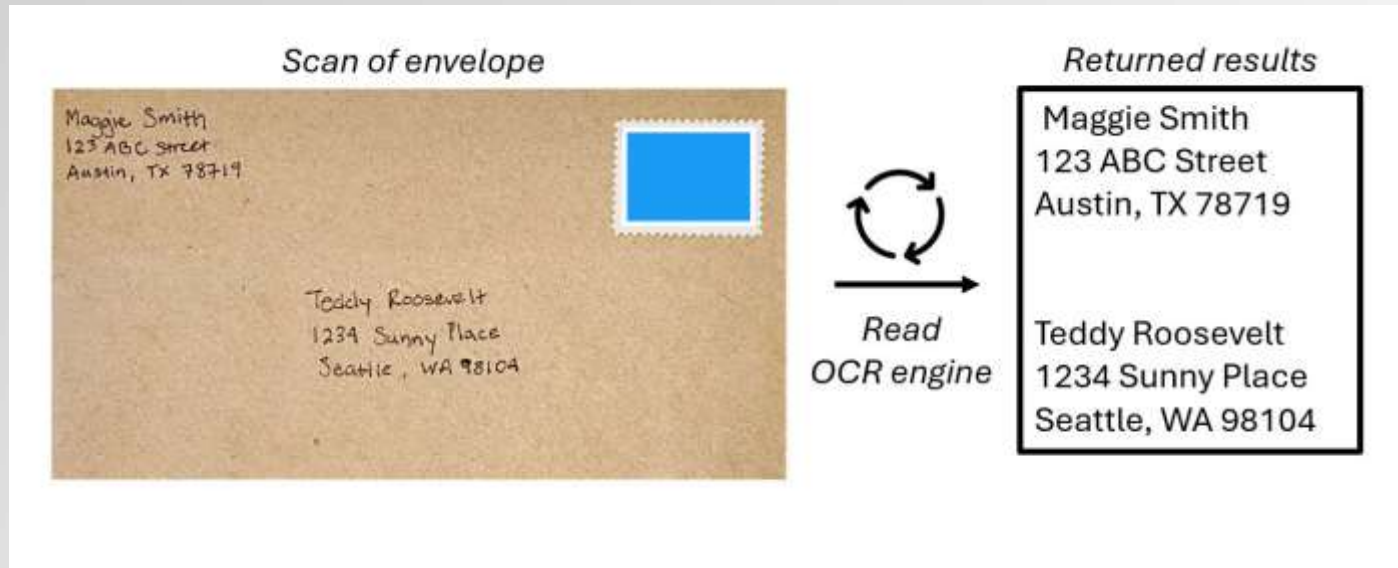
Sub-bab 2

Understand the Extraction of Data from Images



Extraction from Images

Optical Character Recognition (OCR) helps computers recognize that an element in an image contains text. OCR is the foundation of processing text in images and uses machine learning models that are trained to recognize individual shapes as letters, numerals, punctuation, or other elements of text.



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Understand the Extraction of Data from Forms





Extraction Data from Forms

Forms and other documents have text data with ***semantic meaning***. **Semantic meaning refers to the intended meaning or interpretation of words, phrases, or symbols in a given context.** Semantic meaning goes beyond just the literal definition of a word (syntax) and focuses on what the word or sentence actually conveys.

Document intelligence describes AI capabilities that process text and attach semantic meaning to the extracted text. As an extension of optical character recognition (OCR), document intelligence automates the process of extracting and understanding information.



Document Intelligence

Contoso

123 Main Street
Redmond, WA 98052

987-654-3210

6/10/2019 13:59
Sales Associate: Paul

2 Surface Pro 6	\$1,998.00
3 Surface Pen	\$299.97
Sub-Total	\$2,297.97
Tax	\$218.31
Total	\$2,516.28

For example, the address information in on the receipt is saved as a `field name`, `address` and a `value`, `123 Main Street` with coordinates `[4.1, 2.2]`, `[4.3, 2.2]`, `[4.3, 2.4]`, `[4.1, 2.4]`.

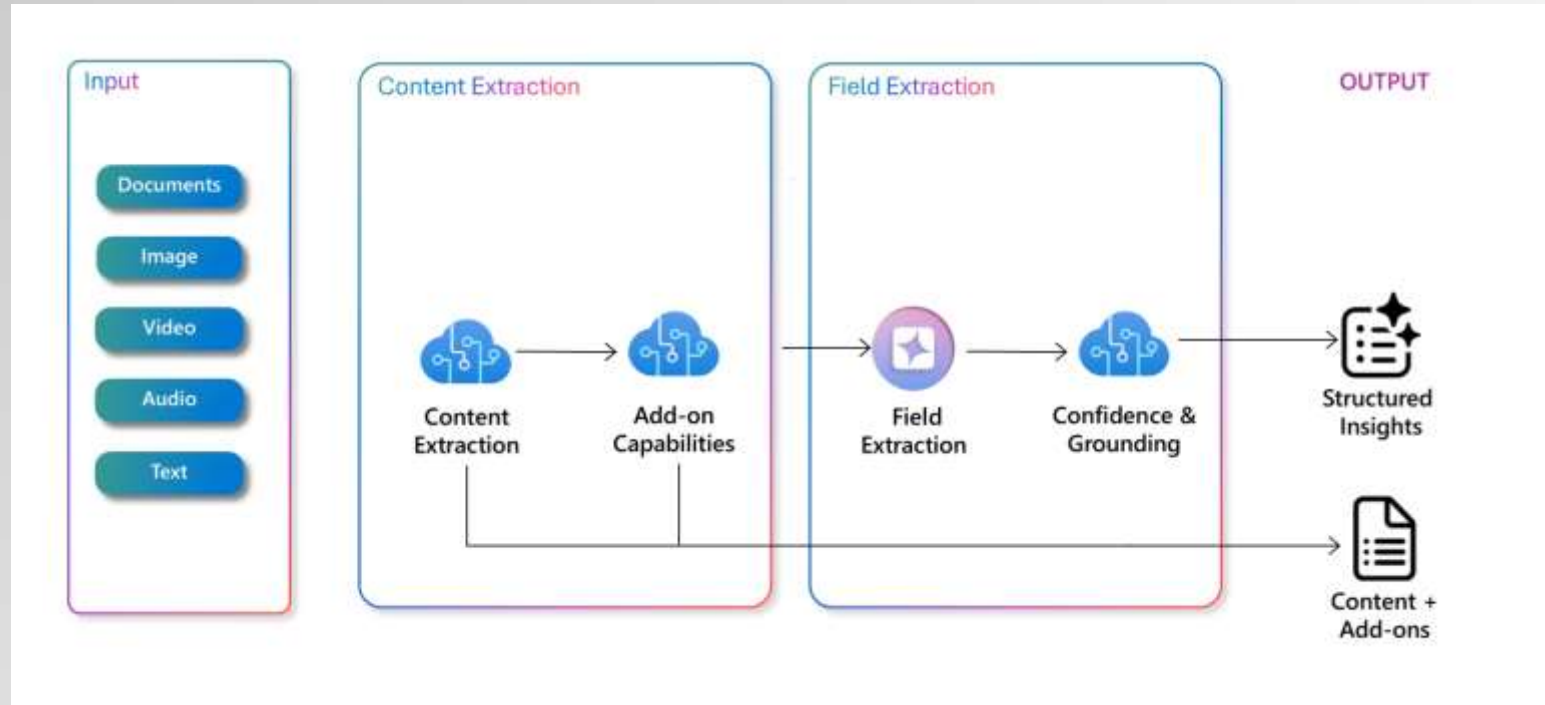
Machine learning models can interpret the data in a document or form because they're trained to recognize patterns in bounding box coordinate locations.

Sub-bab 4

Understand Multimodal Data Extraction



Multimodal Data Extraction



Sub-bab 5

Understand Data Extraction for Knowledge Mining





Data Extraction for Knowledge Mining

Knowledge mining solutions provide automated information extraction from large volumes of often unstructured data. A foundational knowledge mining solution is search, the process of retrieving relevant information from a large dataset in response to a user query. AI-powered information extraction supports improvements in what is searchable in a search index.

The resulting data is serialized as JSON data. The JSON populates the *search index*. The populated search index can be explored through queries.

```
{
  "name": "index",
  "fields": [
    {
      "name": "content", "type": "Edm.String", "analyzer": "standard.lucene", "fields": []
    },
    {
      "name": "keyphrases", "type": "Collection(Edm.String)", "analyzer": "standard.lucene", "fields": []
    },
    {
      "name": "imageTags", "type": "Collection(Edm.String)", "analyzer": "standard.lucene", "fields": []
    }
  ]
}
```

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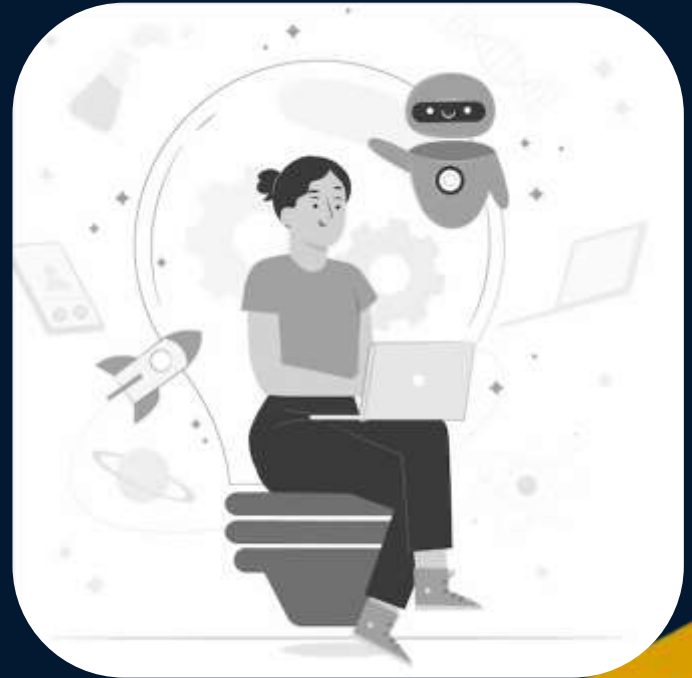


Bab 4

**Get Started with AI-powered
Information Extraction in Azure**

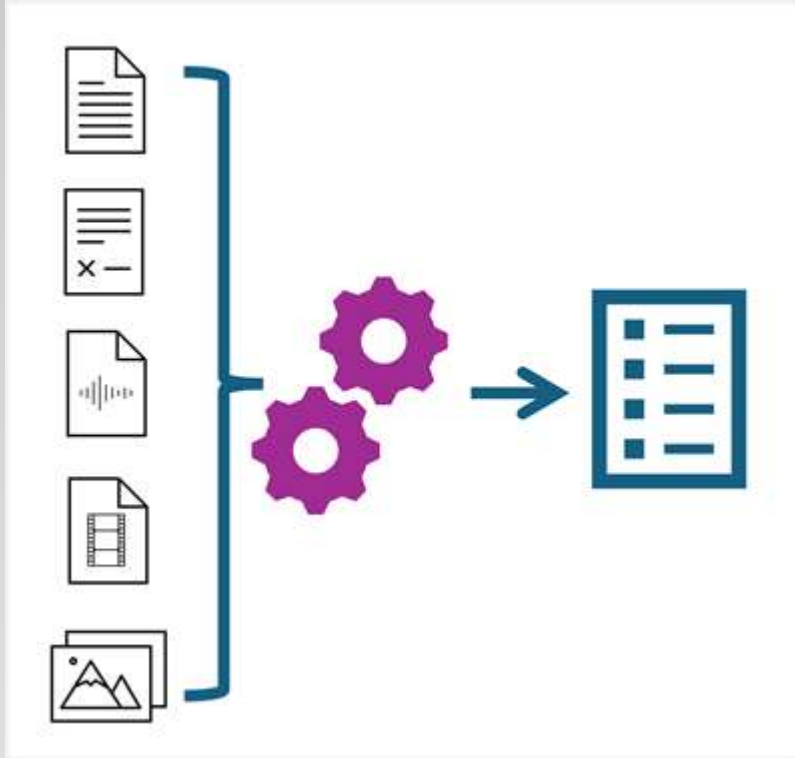
Daftar Isi

1. Introduction
2. Azure AI Services for Information Extraction
3. Extract information with Azure AI Vision
4. Extract multimodal information with Azure AI Content Understanding
5. Extract information from forms with Azure AI Document Intelligence
6. Create a Knowledge Mining solution with Azure AI Search



Bab 4
Sub-bab 1
Introduction

Introduction



AI-powered information extraction and analysis enables organizations to gain actionable insights from data that might otherwise be locked up in documents, images, audio files, or other assets.

Bab 4

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Azure AI Services for Information Extraction



Service

Description



**Azure AI Vision
Image Analysis**

Azure AI Vision Image Analysis enables you to extract insights from images, including the detection and identification of common objects in images, the generation of relevant captions and tags for images, and the extraction of text in images.



**Azure AI Content
Understanding**

Azure AI Content Understanding is a generative AI-based multimodal analysis service that can extract insights from structured documents, images, audio, and video.



**Azure AI Document
Intelligence**

Azure AI Document Intelligence is designed to extract fields and values from digital (or digitized) forms, such as invoices, receipts, purchase orders, and others.



Azure AI Search

Azure AI Search performs AI-assisted indexing in which a pipeline of AI *skills* are used to systematically extract and index information from structured and unstructured content.



You can use each of these services separately, or combine them to build comprehensive solutions for:

- **Data capture:** Intelligently scanning images to capture and store data values. For example, using a cellphone camera to extract contact information from a business card.
- **Business process automation:** Reading data from forms and using it to trigger workflows. For example, extracting cost center and billing information from invoices and routing them to the appropriate accounts-payable department for processing.
- **Meeting summarization and analysis:** Analyzing and summarizing key points from recorded phone conversations or video conference calls. For example, automating note-taking and action assignments for a team meeting.
- **Digital asset management (DAM):** Managing digital assets like images or videos by automatically tagging and indexing them. For example, to create a searchable library of stock photographs.
- **Knowledge Mining:** Extracting key information from structured and unstructured data to be used for further analysis and reporting. For example, compiling census data from scanned records to populate a database.

Bab 4

Sub-bab 3

Extract Information with Azure AI Vision





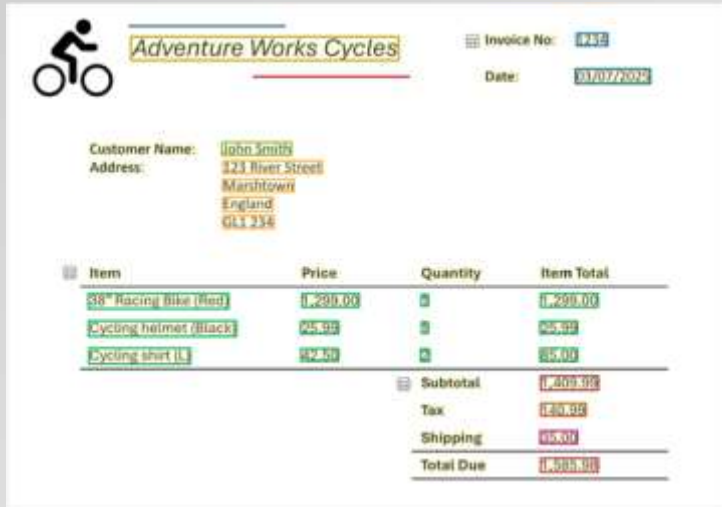
Overview

Azure AI Content Understanding uses state-of-the-art AI models to analyze content in multiple formats, including:

- Text-based forms and documents
- Audio
- Images
- Video

Analyzing Forms and Documents

Azure AI Content Understanding's document analysis capabilities go beyond simple OCR-based text extraction to include schema-based extraction of fields and their values.

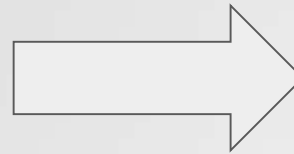


Adventure Works Cycles

Invoice No: 1234
Date: 03/07/2025

Customer Name: John Smith
Address: 123 River Street
Marshalltown
England
GL1 234

Item	Price	Quantity	Item Total
38" Racing Bike (Red)	1,299.00	1	1,299.00
Cycling helmet (Black)	25.99	1	25.99
Cycling shirt (L)	42.50	2	85.00
Subtotal			1,409.99
Tax			140.99
Shipping			35.00
Total Due			1,585.98



For each detected field, the value is extracted from the invoice:

- Vendor name: Adventure Works Cycles
- Invoice number: 1234
- Invoice date: 03/07/2025
- Customer name: John Smith
- Custom address: 123 River Street, Marshalltown, England. GL1 234
- Items:
 - Item 1:
 - Item description: 38" Racing Bike (Red)
 - Unit price: 1299.00
 - Quantity ordered: 1
 - Line item total: 1299.00
 - Item 2:
 - Item description: Cycling helmet (Black)
 - Unit price: 25.99
 - Quantity ordered: 1
 - Line item total: 25.99
 - Item 3:
 - Item description: Cycling shirt (L)
 - Unit price: 42.50
 - Quantity ordered: 2
 - Line item total: 85.00
- Invoice subtotal: 1409.99
- Tax: 140.99
- Shipping Charge: 35.00
- Invoice total: 1585.98



Analyzing Audio

In addition to text-based documents, Azure AI Content Understanding is capable of analyzing audio files to provide transcriptions, summaries, and other key insights.

Now suppose, a caller leaves you the following voice message:

Hi, this is Ava from Contoso.

Just calling to follow up on our meeting last week.

I wanted to let you know that I've run the numbers and I think we can meet your price expectations.

Please call me back on 555-12345 or send me an e-mail at Ava@contoso.com and we'll discuss next steps.

Thanks, bye!

Copy

Using Azure AI Content Understanding to analyze the audio recording and apply your schema produces the following results:

- **Caller:** Ava from Contoso
- **Message summary:** Ava from Contoso called to follow up on a meeting and mentioned that they can meet the price expectations. She requested a callback or an email to discuss next steps.
- **Requested actions:** Call back or send an email to discuss next steps.
- **Callback number:** 555-12345
- **Alternative contact details:** Ava@contoso.com

Analyzing Images and Video

Azure AI Content Understanding supports analysis of images and video to extract information based on a custom schema. For example, you could analyze images of a video conference to extract details of attendance, location, and other information.



- **Location:** Conference room
- **In-person attendees:** 1
- **Remote attendees:** 3
- **Total attendees:** 4

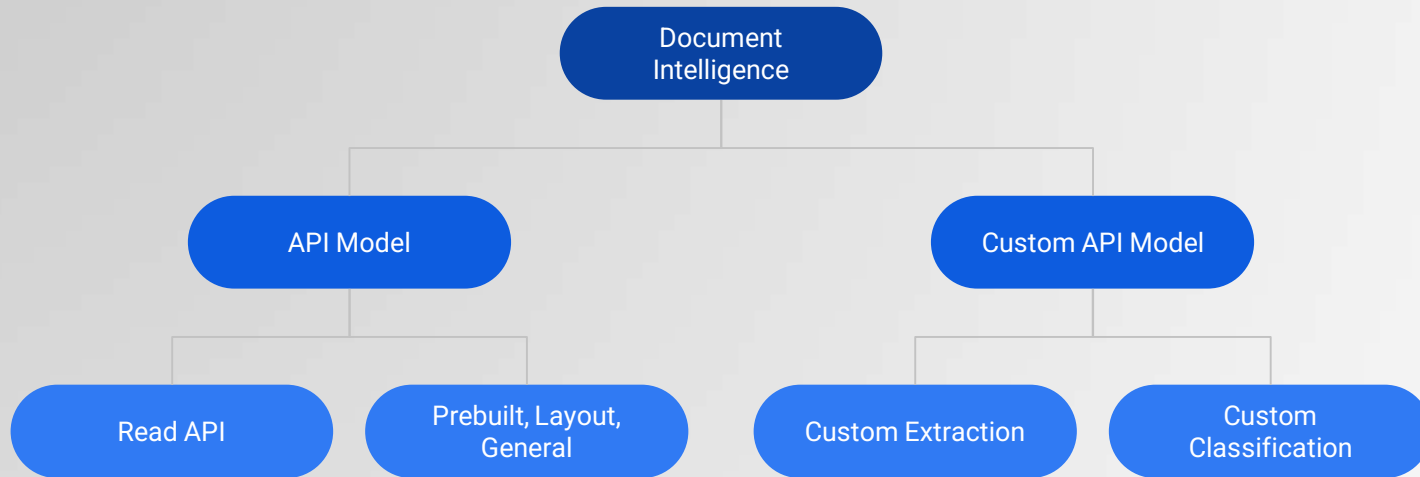
Bab 4

Sub-bab 3

Extract Information from Forms with Azure AI Document Intelligence



Prebuilt and Custom Models



Bab 4

Sub-bab 3

Create Knowledge Mining Solution with Azure AI Search

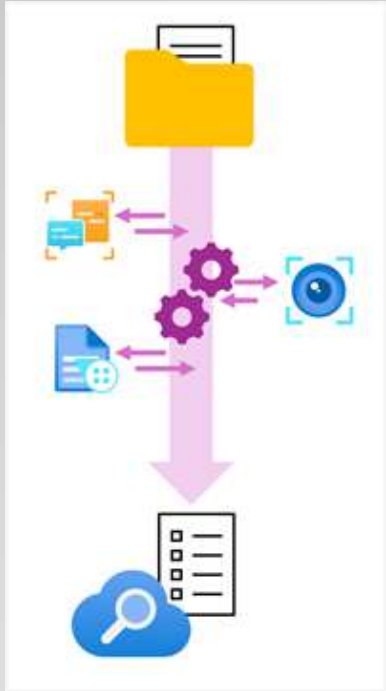




Introduction

Fundamentally, Azure AI Search is a cloud service for indexing and searching data. However, its use of AI *skills* to extract insights from multiple formats of data and the ability to integrate it with other AI services, including Azure AI Vision and Azure AI Document Intelligence make it a powerful platform for building digital asset management and knowledge mining solutions.

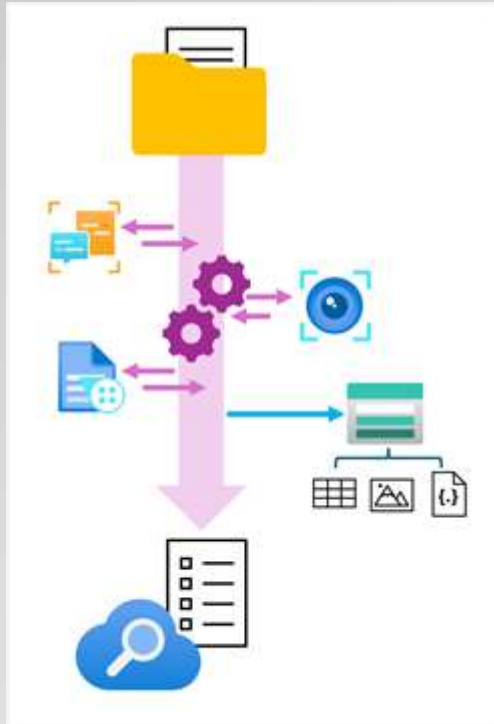
Indexer, Indexes, and Skills



At the heart of an Azure AI Search solution is an *indexer*, which defines a repeatable process to:

1. Ingest data from a *source*, such as an Azure Storage container of documents or a database.
2. *Crack* documents to extract their contents - for example, retrieving the text and image data in a PDF document.
3. Apply a sequence of tasks to retrieve information from the data and generate a hierarchy of *fields* for the index. Some fields are core attributes of the source data (for example document file names and last saved dates), while others are generated by using AI *skills*. For example:
 - Using Azure AI Vision services to generate *tags* and *captions* for images.
 - Using Azure AI Language services to derive fields for *sentiment* or *named entities*.
 - Using Azure AI Document Intelligence to extract field values from forms.
4. Persisting the extracted fields as an *index*.

Persisting Extracted Data to a Knowledge Store



As well as creating a searchable index, Azure AI Search can persist the extracted data assets to a *knowledge store* in Azure Storage.

The indexer can save the following kinds of asset in a knowledge store:

- Tables of field values.
- Images extracted from documents.
- JSON documents representing data structures; which can be complex hierarchies of fields and values.

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Thank you!

Implementing Partner:

